FET physics

Answers and worked solutions · Day 15 · Sat 7 Nov · pack E9

5. Practice

From your course files

WS7 Q3 (course question).

This is Worked Example 2, where the full working is laid out.

a. −2 V lies between $V_P = -4.5$ V and 0, so Shockley applies:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 6\left(1 - \frac{-2}{-4.5}\right)^2 = 6(0.5556)^2 = \mathbf{1.852\ mA}$$

b. $$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right) = -4.5\left(1 - \sqrt{\tfrac{3}{6}}\right) = \mathbf{-1.318\ V}$$

Check: 3 mA is $I_{DSS}/2$, so this is the "$0.3V_P$" point ($0.3 \times -4.5 = -1.35$ V). The two values agree.

WS7 Q6 (course question).

Hint: $V_{GS} = V_P/2$, so $I_D = I_{DSS}/4$.

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \Rightarrow 3\ \text{mA} = I_{DSS}\left(1 - \frac{-3}{-6}\right)^2 = 0.25\, I_{DSS}$$

$$I_{DSS} = \frac{3\ \text{mA}}{0.25} = \mathbf{12\ mA}$$

WS7 Q5 (course question).

Use the four-point method (2.2):

V_{GS} (V)	0	$0.3V_P = -1.2$	$V_P/2 = -2$	−3 (extra point, Shockley)	$V_P = -4$
I_D (mA)	$I_{DSS} = 12$	$I_{DSS}/2 = 6$	$I_{DSS}/4 = 3$	0.75	0

Plot the points on I_D (mA) against V_{GS} (V) and draw a smooth concave-up curve that meets the V_{GS} axis tangentially at −4 V. Label I_{DSS} and V_P.

Notes on the official solution:

- It writes "Given $I_{DSS} = 6$ mA", which is a typo. All of its numbers use 12 mA, which is correct.
- The exact V_{GS} for $I_{DSS}/2$ is $-4(1 - \sqrt{0.5}) = -1.172$ V. The −1.2 V is the shorthand $0.3V_P$ value, which is fine for a sketch.

WS7 Q1 (course question).

Assume an n-channel JFET, so $V_P = -6$ V. Use method 2.3: compute the saturation current for each 1 V step of V_{GS} with Shockley, $I_D = 12(1 - V_{GS}/(-6))^2$ mA.

V_{GS} (V)	0	−1	−2	−3	−4	−5	−6 (V_P)
I_D (mA)	12 (I_{DSS})	8.33	5.33	3	1.33	0.333	0

Draw each as a horizontal line on the I_D–V_{DS} axes, starting at its knee $V_{DS} = V_{GS} - V_P$ (6, 5, 4, 3, 2, 1 V), with a curve back to the origin for the ohmic region. Join the knees with a dashed pinch-off locus. The $V_{GS} = -6$ V curve lies on the V_{DS} axis. Note how the curves crowd together as V_{GS} approaches V_P.

The official solution writes "$3V_{GS} = V_P$", which is a typo for "$V_{GS} = V_P$".

WS7 Q7 (course question).

"$V_{GS} = -V_P$ to 1 V" means from $V_{GS} = V_P = -8$ V up to +1 V (the sheet's sign is a slip: $-V_P$ would be +8 V).

Shockley, $I_D = 12(1 - V_{GS}/(-8))^2$ mA:

V_{GS} (V)	−8 (V_P)	−6	−4 ($V_P/2$)	−2.4 ($0.3V_P$)	0	+1
I_D (mA)	0	0.75	3	6 ($I_{DSS}/2$)	12 (I_{DSS})	**15.19**

- **Transfer curve:** plot these points and draw a smooth concave-up curve. It continues to the right of the I_D axis and rises above I_{DSS} for $V_{GS} > 0$ (enhancement mode).
- **Drain curves:** one flat saturation line per V_{GS} at the current in the table, with the knee at $V_{DS} = V_{GS} - V_P$, and the +1 V curve above the I_{DSS} curve.

Notes on the official solution:

- Its point list stops at 0 V. Add $V_{GS} = +1$ V: $I_D = 12(1 + 1/8)^2 = 15.19$ mA, so the curve continues above I_{DSS}.
- The exact V_{GS} for $I_{DSS}/2$ is −2.343 V. The −2.4 V is the shorthand $0.3V_P$ value.

WS7 Q2 (course question).

Read the flat (saturation) part of each curve. "$V_{DS} > V_P$" means V_{DS} beyond the pinch-off value, $|V_P| = 4$ V.

- $V_{GS} = 0$ V: $I_D = 8$ mA (I_{DSS}).
- $V_{GS} = -1$ V: $I_D = 4.5$ mA.

Difference: $8 - 4.5 = \mathbf{3.5\ mA}$.

Check: Fig. 1 is the slide-10 device. Shockley with $I_{DSS} = 8$ mA, $V_P = -4$ V gives $8(1 - 0.25)^2 = 4.5$ mA at −1 V. It matches.

WS7 Q4 (course question).

Read the scales first: 1 mA per vertical division, 1 V per horizontal division. So $V_{DS} = 10$ V is the right-hand edge of the grid. $I_{DSS} \approx 9$ mA and $V_P \approx -3$ V.

$V_{GS} = -0.7$ V lies between the −0.5 V curve (≈ 6.5 mA) and the −1 V curve (≈ 4 mA). The curves are not evenly spaced, so interpolate a little below the midpoint: $I_D \approx 5.5$ **mA** (the official answer).

Shockley cross-check with the figure's own values: $9(1 - 0.7/3)^2 = \mathbf{5.29\ mA}$. Both are acceptable, because a graph reading depends on how you interpolate between the −0.5 V and −1 V curves.

The official solution text writes "$V_{GS} = 0.7$ V", which is a typo for −0.7 V.

New questions

Q1 [3 marks]

a. $V_P > 0$ for p-channel, and +2 V lies between 0 and +6 V, so Shockley applies:

$$I_D = 6\left(1 - \frac{2}{6}\right)^2 = 6(0.6667)^2 = \mathbf{2.67\ mA}$$

b. 1.5 mA is $I_{DSS}/4$, so expect $V_P/2$:

$$V_{GS} = 6\left(1 - \sqrt{1.5/6}\right) = 6(0.5) = \mathbf{+3\ V}$$

c. Conventional current flows from **source to drain**, so it leaves by the drain terminal. V_{DS} is **negative**. The equation is the same as for n-channel, but with V_P and V_{GS} positive.

Q2 [4 marks]

a. $k = \dfrac{9\ \text{mA}}{(8-2)^2} = \dfrac{9}{36} = \mathbf{0.25\ mA/V^2}$

b. $I_D(5\ \text{V}) = 0.25(5-2)^2 = \mathbf{2.25\ mA}$. At 1.5 V the gate is below $V_T = 2$ V, so there is no channel and $I_D = \mathbf{0}$.

c. $V_{GS} = V_T + \sqrt{I_D/k} = 2 + \sqrt{4/0.25} = 2 + 4 = \mathbf{6\ V}$. The other root, $2 - 4 = -2$ V, is below V_T, so it is rejected.

d. $V_{DS(\text{sat})} = V_{GS} - V_T = 6 - 2 = \mathbf{4\ V}$. Saturation needs $V_{DS} \geq 4$ V.

Q3 [5 marks]

a.

- $I_D(-1) = 8\left(1 - \frac{-1}{-3}\right)^2 = 8(0.6667)^2 = \mathbf{3.56\ mA}$
- $I_D(-1.5) = 8(0.5)^2 = \mathbf{2.00\ mA}$ (this is the $V_P/2$ point, $I_{DSS}/4$)
- $I_D(+1) = 8\left(1 - \frac{1}{-3}\right)^2 = 8(1.3333)^2 = \mathbf{14.22\ mA}$

b. $V_{GS} < 0$ (−1 V and −1.5 V) is **depletion** mode, with $I_D < I_{DSS}$. $V_{GS} > 0$ (+1 V) is **enhancement** mode, with $I_D > I_{DSS}$.

c. $V_{GS} = -3(1 - \sqrt{4/8}) = \mathbf{-0.879\ V}$. The shorthand gives $0.3V_P = -0.9$ V, about 0.02 V away.

Q4 [8 marks] (chained)

a. Take the ratio of the two Shockley equations, so that I_{DSS} cancels:

$$\sqrt{\frac{4.5}{0.5}} = 3 = \frac{1 - (-1)/V_P}{1 - (-3)/V_P} = \frac{1+u}{1+3u}, \quad u = \frac{1}{V_P}$$

$$1 + u = 3 + 9u \ \Rightarrow\ u = -0.25 \ \Rightarrow\ \mathbf{V_P = -4\ V}$$

b. $I_{DSS} = \dfrac{4.5}{(1 - (-1)/(-4))^2} = \dfrac{4.5}{0.5625} = \mathbf{8\ mA}$

Check with the second point: $8(1 - 0.75)^2 = 0.5$ mA. It matches.

c. $I_D = 8(1 - 0.5)^2 = \mathbf{2\ mA}$. This is the $V_P/2 \to I_{DSS}/4$ point. (This is the slide-10 device.)

d. $V_{DS(\text{sat})} = V_{GS} - V_P = -2 - (-4) = \mathbf{2\ V}$

e. $r_d = \dfrac{r_o}{(1 - V_{GS}/V_P)^2} = \dfrac{250}{(0.5)^2} = \mathbf{1\ k\Omega}$

The resistance rises as V_{GS} goes more negative. That is the voltage-controlled-resistor behaviour.

Q5 [6 marks]

a. $V_{GS,n} = V_i$ and $V_{GS,p} = V_i - 5$.

V_i	$V_{GS,n}$	nMOS	$V_{GS,p}$	pMOS	V_o
0	0 V	off (< 1 V)	−5 V	on (< −1 V)	5 V
2.5 V	2.5 V	on	−2.5 V	on	transition region
5 V	5 V	on	0 V	off	0 V

At 2.5 V both transistors conduct. The output is in the transition (amplifier) region, and a current flows from V_{DD} to ground. In either logic state one device is off, so there is no static current.

b. $NM_L = 2.1 - 0.1 = \mathbf{2.0\ V}$. $NM_H = 4.9 - 2.9 = \mathbf{2.0\ V}$. The ideal inverter has $V_{IL} = V_{IH} = 2.5$ V, $V_{OL} = 0$ and $V_{OH} = 5$ V, giving 2.5 V each. The real margins are smaller because the transition region has finite width.

c. $t_P = \frac{1}{2}(12 + 18) = \mathbf{15\ ns}$

Q6 [6 marks] (hard, chained)

a. $k = \dfrac{5}{(-6.5 - (-1.5))^2} = \dfrac{5}{25} = \mathbf{0.2\ mA/V^2}$

b. $|I_D| = 0.2(-4 + 1.5)^2 = 0.2(6.25) = \mathbf{1.25\ mA}$

c. $(V_{GS} + 1.5)^2 = 1.8/0.2 = 9$, so $V_{GS} + 1.5 = \pm 3$. The roots are $V_{GS} = -4.5$ V and $V_{GS} = +1.5$ V. A pMOS is on only for $V_{GS} < V_T = -1.5$ V, so $\mathbf{V_{GS} = -4.5\ V}$. The +1.5 V root is not physical for this device. At that point $V_{DS(\text{sat})} = V_{GS} - V_T = -3$ V, so saturation needs $V_{DS} \le -3$ V.

d. $V_{GS,p} = V_i - V_{DD} = 0.5 - 5 = \mathbf{-4.5\ V}$, which is the part (c) point, so the saturation current is $\mathbf{1.8\ mA}$. The nMOS has $V_{GS} = 0.5\ \text{V} < 1.5$ V, so it is **off**. No

current path exists, the pMOS pulls the output to V_{DD}, and $V_o \approx 5$ V (logic 1). Because no current flows, the pMOS actually sits in its ohmic region with $V_{DS} \approx 0$. The 1.8 mA is only what it *could* supply, e.g. to charge a load capacitance.

6. Quick recall check

Question

1. $V_{GS} = 0$ V and $V_{DS} \geq |V_P|$ (saturation). It is the maximum JFET current.
2. $I_{DSS}/4$ (the $V_P/2$ point). $V_{GS} \approx 0.3V_P = -1.5$ V (exactly −1.46 V).
3. The enhancement MOSFET: $I_D = k(V_{GS} - V_T)^2$ for $V_{GS} > V_T$, and 0 otherwise.
4. $V_{GS} = 0 - 5 = -5$ V. It is **on**, and the output is high.
5. $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$.